

Quiz 1:

HS7.301 Science, Technology and Society

Max Marks: 30

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**What does the passage below tell you about the nature of scientific knowledge making?
Analyse using frameworks discussed in class.**

Ancient civilizations, including the Egyptians, Greeks, and Romans, long pondered the mysterious origins of disease and often attributed illness to supernatural forces or environmental influences. Early physicians such as Hippocrates and Galen sought more naturalistic explanations and proposed theories centred on bodily processes, even as they speculated about invisible environmental agents as possible causes of disease. Hippocrates, often referred to as the “Father of Medicine,” developed the theory of the four humors, according to which health depended on the balance of four bodily fluids—blood, phlegm, black bile, and yellow bile. Disease, in this view, resulted from an imbalance among these humors.

Alongside humoral theory emerged the miasma theory, also associated with Hippocratic thought. Derived from the Greek word for pollution or “bad air,” the miasma theory held that diseases were caused by noxious vapours arising from decomposing organic matter such as plants or animals. These vapours, identified by their foul odour, were believed to contaminate the air, and those who inhaled this polluted atmosphere were thought to fall ill. It was observed, for example, that colds and flu-like illnesses were more common during cold and wet conditions, reinforcing the belief that environmental factors played a decisive role in disease.

During the early modern period, alternative explanations of disease transmission began to emerge. Girolamo Fracastoro, a sixteenth-century Italian physician, was among the first to use the term “contagion” to explain how illnesses spread. His theory of “seeds of disease” proposed that diseases could be transmitted through direct contact, indirect contact via objects, or even across distances without physical contact. Although influential, Fracastoro’s ideas did not immediately displace miasmatic explanations. In the nineteenth century, Justus von Liebig, a German chemist and pioneer of organic chemistry, advanced another influential view, arguing that disease arose in the blood due to chemical processes triggered by the decay of organic matter. He described the blood as the body’s “chemical factory,” further reinforcing non-microbial explanations of disease.

A major shift became possible with the discovery of the microscope. Antonie van Leeuwenhoek, a Dutch scientist, developed the first single-lens microscopes in the late seventeenth century, enabling the observation of microorganisms invisible to the naked eye. His work laid the foundations for microbiology as a scientific field, although the significance of microorganisms for disease was not immediately recognised. Over time, microbiology developed into a discipline concerned with the study of microorganisms and the development of medicines to prevent and treat disease. Two core principles gradually emerged within bacteriology: that diseases originate from microbes and that diseases are transmitted from person to person through the spread of these microscopic organisms.

*In the nineteenth century, John Snow played a crucial role in challenging miasma theory. Snow rejected the idea that all diseases were transmitted through contaminated air and instead argued that cholera was spread through contaminated water. In his 1849 work *On the Mode of Communication of Cholera*, he demonstrated that cholera entered the body through ingestion and affected the intestines rather than the lungs. During the 1854 cholera outbreak*

in London, Snow traced the source of infection to a water pump on Broad Street that drew contaminated water from the Thames. His recommendation that people boil their drinking water later became a cornerstone of public health practice.

At roughly the same time, Ignaz Semmelweis, a Hungarian physician, introduced antiseptic protocols in hospital wards. He discovered that physicians who moved directly from autopsies to maternity wards transmitted disease to patients, whereas midwives who did not handle cadavers had far lower mortality rates. Semmelweis advocated handwashing between handling cadavers and attending patients, a practice that significantly reduced cases of puerperal sepsis, also known as childbed fever. Building on similar insights, Joseph Lister later advanced antiseptic surgery by demonstrating that microbes caused wound decay. Lister promoted the sterilisation of surgical instruments and the washing of surgeons' hands, leading to dramatic reductions in surgical mortality.

The germ theory of disease was formally articulated by Louis Pasteur, who presented the theory to the French Academy of Medicine in 1878. Pasteur's work began earlier, in 1857, when he was asked to investigate why beer and wine spoiled. He discovered that microorganisms were responsible for fermentation and spoilage and proposed heating liquids to kill these microbes, a process later known as pasteurization. Pasteur rejected miasma and spontaneous generation, arguing instead that specific microorganisms caused specific infectious diseases. His development of attenuated vaccines, including the rabies vaccine, demonstrated the practical power of this new framework. Along with colleagues Jules Joubert and Charles Chamberland, Pasteur articulated key principles of germ theory: that specific microbes cause specific conditions, that microbes are responsible for disease, fermentation, and putrefaction, and that identifying particular microbes makes vaccination possible.

Robert Koch further strengthened germ theory through his work in medical microbiology. He developed Koch's postulates, a set of criteria for establishing a causal relationship between a microorganism and a disease. Using these principles, Koch identified the agents responsible for anthrax, tuberculosis, and cholera. His use of staining techniques, including what later became Gram staining, allowed scientists to visualise and differentiate bacteria.

The acceptance of germ theory transformed public health. During the nineteenth and twentieth centuries, municipalities introduced large-scale measures such as sewer systems, water treatment, waste management, street cleaning, and food safety regulations. At the same time, cleanliness campaigns promoted personal hygiene, encouraging regular washing, clean clothing, and proper waste disposal. These developments also gave rise to "scientific housekeeping" or home economics, which emphasised domestic cleanliness as a means of combating germs. The impact of germ theory extended into modern medicine with the discovery of antibiotics, beginning with Alexander Fleming's discovery of penicillin. By revealing the microbial causes of disease, germ theory has fundamentally reshaped medicine and public health, saving countless lives that would otherwise have been lost to invisible pathogens.

Thomas Kuhn's framework of scientific change provides a useful lens for analysing the historical transition from miasma and humoral theories of disease to the germ theory outlined in the passage. Rather than presenting scientific progress as a steady accumulation of knowledge, Kuhn argues that science advances through periods of "normal science" punctuated by crises and paradigm shifts. The passage illustrates each of these stages clearly. For long periods, medical knowledge operated under what Kuhn would describe as a dominant paradigm. Humoral theory and miasma theory together structured medical research, training, and public health practice. Physicians shared common assumptions about the causes of disease, focusing on environmental conditions, bodily fluids, smells, and climate. This

corresponds to Kuhn's notion of normal science, in which scientists do not question foundational assumptions but instead engage in puzzle-solving within an accepted framework. Medical education, urban sanitation policies, and clinical practices were all oriented toward refining and applying these ideas rather than testing their validity.

Within this period of normal science, anomalies did appear but were initially accommodated rather than treated as fundamental problems. Observations such as the uneven spread of cholera, Fracastoro's notion of contagion, and early microscopic observations did not immediately destabilise the prevailing paradigm. As Kuhn suggests, anomalies are often explained away through auxiliary hypotheses or treated as special cases. In the passage, Snow's early findings linking cholera to water were acknowledged but interpreted as local irregularities, while miasmatic explanations remained dominant. This reflects Kuhn's argument that paradigms are resilient and not easily abandoned in the face of contradictory evidence.

Over time, however, anomalies accumulated and became harder to ignore. Advances in microscopy, laboratory experimentation, and epidemiological methods increasingly challenged miasmatic explanations. Disagreements within the medical community intensified, particularly over what counted as legitimate evidence. Laboratory-based bacteriology conflicted with environmental observation and statistical correlations, indicating a breakdown of shared standards. This stage corresponds to Kuhn's description of a crisis, in which confidence in the existing paradigm weakens and competing approaches proliferate.

The emergence of germ theory represents a scientific revolution or paradigm shift in Kuhn's sense. Pasteur and Koch did not merely add new facts to existing theories but redefined the fundamental concepts of disease. Illness was no longer understood primarily in terms of environmental conditions or bodily imbalances but as the result of infection by specific microorganisms. New methods, such as culturing bacteria and applying Koch's postulates, replaced earlier observational practices. Problems that were once central—such as foul smells or atmospheric conditions—lost their explanatory significance, while previously puzzling anomalies became intelligible within the new framework.

The passage also illustrates Kuhn's idea of incommensurability. The miasma and germ paradigms differed not only in their explanations but in their standards of proof, methods, and language. Concepts like "bad air" and "infection" were not directly translatable, and practitioners trained in one framework often found the other unintuitive or incomplete. Resistance from physicians who emphasised social and environmental factors reflects Kuhn's claim that paradigm choice is not determined solely by empirical evidence but also by values, training, and professional commitments.

Finally, the institutional changes described in the passage—rewritten textbooks, new curricula, laboratory-based training, and public health reforms—demonstrate how paradigm shifts become stabilised. Once germ theory became dominant, it defined a new period of normal science, within which microbiology, antibiotics, and molecular biology developed. From a Kuhnian perspective, this transition shows that scientific progress involves discontinuities and reorganisations of knowledge rather than linear advancement toward truth.

2. Why is the process of resolving controversies so central in the making of scientific knowledge? Explain using examples, how the resolving of controversies involves both the 'social' and the 'scientific'.